

# Singular anharmonicities and the analytic continued fractions

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The potential  $V(r) = \mu^2 r^2 + \lambda^2 r^{-4}$  is investigated and its bound states are constructed by a generalized Hill-determinant method. It is shown that the binding energy  $E$  and another free parameter  $\kappa$  (in the underlying Laurent-series representation of the wave function) may be obtained from a coupled pair of Hill-determinant equations.

## I. INTRODUCTION

The singularly anharmonic potential

$$V(r) = \mu^2 r^2 + \lambda^2 r^{-4} \quad (1.1)$$

is a superposition of the two forces such that in both the extreme cases  $\mu = 0$  and  $\lambda = 0$ , the complete three-dimensional Schrödinger equation becomes solvable exactly.<sup>1</sup> Nevertheless, an infinitesimal transition to  $\mu \neq 0$  and  $\lambda \neq 0$  represents such a drastic change of the interaction that a perturbation theory fails to give any estimates: the coupling  $\mu \neq 0$  introduces a confinement and  $\lambda \neq 0$  also represents a singular perturbation. Methodically, such a situation simulates the difficulties encountered, e.g., in the contemporary field theory.<sup>2</sup>

Our present intention is to develop a consistent expansion and solution method. Technically, we are inspired by the paper of Singh *et al.*<sup>3</sup> where a sextic anharmonicity has been treated by the nonperturbative means, based on the use of analytic continued fractions. We shall arrive here at a similar "non-numerical" solution of the eigenvalue problem, which may be interpreted as a "natural" resummation of the divergent perturbative expansions.

Phenomenologically, the form of our potential (1.1) represents an independent motivation for an interest in the corresponding radial Schrödinger equation

$$\left( -\frac{d^2}{dr^2} + \frac{l(l+1)}{r^2} + \mu^2 r^2 + \frac{\lambda^2}{r^4} - E \right) \Psi(r) = 0, \quad l = 0, 1, \dots \quad (1.2)$$

Indeed, the harmonic component of the force is a common approximation verified in the various realistic situations. From a physical point of view, the effects of a finite size or correlations of the interacting objects are often important just at the short distances (cf. the phenomenological nucleon-nucleon interactions, etc.). Thus, in a purely phenomenological sense, a singular repulsion seems to be a very useful form of an anharmonicity.

## II. NUMERICAL METHODS

A naive treatment of  $\lambda^2 r^{-4}$  as a small perturbation leads immediately to contradictions: for  $l = 0$ , the first nontrivial correction to the energy (i.e., the matrix element) diverges as an integral in the origin,

$$\langle n | V | n \rangle \sim \int_{\epsilon}^{\infty} \langle r | n \rangle^2 r^{-4} r^2 dr \sim \frac{1}{\epsilon}, \quad \epsilon \rightarrow 0.$$

A perturbative interpretation of  $\mu$  also leads to difficulties: the  $\mu = 0$  case does not possess the discrete spectrum and may be characterized by an abrupt disappearance of the confinement.

In a purely numerical framework, the differential equation (1.2) is singular in the origin: the initial boundary conditions at  $r = 0$  are ambiguous and a nontrivial modification of the standard techniques<sup>4,5</sup> must be employed.

The more reliable approach may be expected to follow from the various types of the variational *Ansätze*. For a suitably modified standard basis (cf., e.g., Ref. 5), an *a priori* slow rate of convergence with respect to an increasing cutoff dimension may be accelerated by the various techniques (cf., e.g., Ref. 6). Alternatively, an application of the Lanczos algorithm<sup>7</sup> may permit one to choose an "optimal" initial guess of  $|0\rangle$  and to construct the basis  $\{|n\rangle\}$  and improve the results in a systematic way.

After a brief inspection of the differential equation (1.2), we may arrive at the estimates

$$\psi(r) \sim \exp(-\lambda r^{-1}), \quad r \approx 0, \quad (2.1)$$

and

$$\psi(r) \sim \exp(-\frac{1}{2}\mu r^2), \quad r \rightarrow \infty, \quad (2.2)$$

of the respective threshold and asymptotic physical behavior of the normalizable bound-state solutions. In this way, the "natural" Lanczos initial state  $\langle r | 0 \rangle$  has the form

$$\langle r | 0 \rangle = r^{\kappa} \exp(-\lambda r^{-1} - \frac{1}{2}\mu r^2), \quad (2.3)$$

where  $\kappa$  is some new free parameter.

The formalism of Lanczos is based on a subsequent orthogonalization of the states,<sup>8</sup>  $|n\rangle \sim H|n-1\rangle + \dots$ . After an appropriate variational truncation, it leads to the approximate wave functions of the type  $\sum_n \langle r | n \rangle \times c_n$ , with

$$\langle r | n \rangle \sim r^{\kappa} \times \text{constant} \times \langle r | 0 \rangle \times \text{a polynomial}. \quad (2.4)$$

This will be employed below as a methodical inspiration.

## III. HILL DETERMINANTS AND THE ANALYTIC CONTINUED FRACTIONS

Mathematically, the main shortcoming of the variational estimates (2.4) lies in an unclear role played by the trial parameter  $\kappa$ . Indeed, in the case of the regular potentials, this is not a free parameter ( $\kappa = l + 1$  specifies solutions regular in the origin). Here, we shall interpret Eq. (2.4) as a tentative *Ansatz* and analyze its consequences in a non-numerical manner.

For the sake of simplicity we shall omit the parity-violating factor  $\exp(-\lambda r^{-1})$  and start from the formula

$$\psi(r) = \exp\left(-\frac{1}{2}\mu r^2\right) \sum_{n=-\infty}^{\infty} p_n r^{2n+\kappa}. \quad (3.1)$$

Its insertion in the Schrödinger equation (1.2) converts this ordinary differential equation of second order into the relations

$$\begin{aligned} A_{n+1}p_n + B_{n+1}p_{n+1} + C_{n+1}p_{n+2} &= 0, \\ A_{n+1} &= (4n+2\kappa+1)\mu - E, \\ B_{n+1} &= -(2n+\kappa+1)(2n+\kappa+2) + l(l+1), \\ C_{n+1} &= \lambda^2, \quad n = \dots, -1, 0, 1, \dots \end{aligned} \quad (3.2)$$

They may be treated as recurrences or as a difference equation of the second order. Of course, our first problem lies now in a rigorous determination of the suitable initial or boundary conditions pertaining to Eq. (3.2).

In the first step of our considerations, an intuitive “variational” truncation of the doubly infinite system of equations (3.2) may be introduced,

$$\det \mathcal{H}_{-M,N} = 0, \quad \mathcal{H}_{-M,N} = \begin{pmatrix} B_{-M} & C_{-M} & & & \\ & \ddots & & & \\ & & A_{k-M} & B_{k-M} & C_{k-M} \\ & & \ddots & & \\ & & & A_N & B_N \end{pmatrix}, \quad M, N \gg 1. \quad (3.3)$$

Later on (in Sec. IV) this equation will become a part of our rigorous construction. Here, let us start its analysis in the spirit of Ref. 3, i.e., postulating it as a source of the “quasi-variational” energy approximants of an *a priori* unknown validity.

When we put

$$f_n = p_n/p_{n-1}, \quad N \gg n > 0, \quad (3.4)$$

and

$$g_m = p_{-m}/p_{1-m}, \quad M \gg m > 0, \quad (3.5)$$

in Eq. (3.2), we may interpret the quantities  $f_n$  and  $g_m$  as the finite continued-fractional approximants.<sup>9</sup> Indeed, the formal initial values  $f_{N+1} = 0$  and  $g_{M+1} = 0$  are to be employed in the corresponding recurrences (3.2), i.e.,

$$f_n = -A_n/(B_n + C_n f_{n+1}), \quad n = 1, 2, \dots, N, \quad N \rightarrow \infty, \quad (3.6)$$

and

$$\begin{aligned} g_m &= -C_{-m}/(B_{-m} + A_{-m} g_{m+1}), \\ m &= 1, 2, \dots, M, \quad M \rightarrow \infty. \end{aligned} \quad (3.7)$$

In this way, an infinite-dimensional limit  $M, N \rightarrow \infty$  of the Hill-determinant condition (3.3) becomes equivalent to the continued-fractional condition

$$A_0 g_1 + B_0 + C_0 f_1 = 0. \quad (3.8)$$

The proof of this statement is easy—the triplet of Eqs. (3.6)–(3.8) is precisely equivalent to (is a mere transcription of)

the original set (3.2). Moreover, we have also the following mathematical result at our disposal.

**Lemma 1:** Continued fractions  $f_n$  and  $g_m$  are convergent.

**Proof:** For  $n \gg 1$ , an almost  $n$ -independent mapping  $f_{n+1} \rightarrow f_n$  is defined by Eq. (3.6),

$$\begin{aligned} f_n &= \mu/(n_0 - \lambda^2 f_{n+1}/(4n_0)) \\ &+ \text{corrections}, \quad n = O(n_0) \gg 1. \end{aligned} \quad (3.9)$$

In accord with Fig. 1, it has a simple geometric interpretation. The sequence  $f_{n_0}, f_{n_0-1}, \dots$  will accumulate near a stable fixed point  $\varphi = \varphi(n)$  defined by the quadratic equation

$$\varphi(n) = \mu/(n - \lambda^2 \varphi(n)/(4n)), \quad n = O(n_0) \gg 1. \quad (3.10)$$

Thus, from an arbitrary initial value of  $f_{n_1} = O(1)$  at some sufficiently large index  $n_1 \gg n_0$ , we get

$$f_n = \mu/n + \text{corrections}, \quad n = O(n_0) \gg 1. \quad (3.11)$$

*Mutatis mutandis*, the asymptotic form

$$\begin{aligned} g_m &= \lambda^2/(4m_0^2 + 4\mu m_0 g_{m+1}) \\ &+ \text{corrections}, \quad m = O(m_0) \gg 1, \end{aligned} \quad (3.12)$$

of the “conjugate” mapping (3.7) leads to an analog of Eq. (3.11),

$$g_m = \lambda^2/(4m^2) + \text{corrections}, \quad m = O(m_0) \gg 1. \quad (3.13)$$

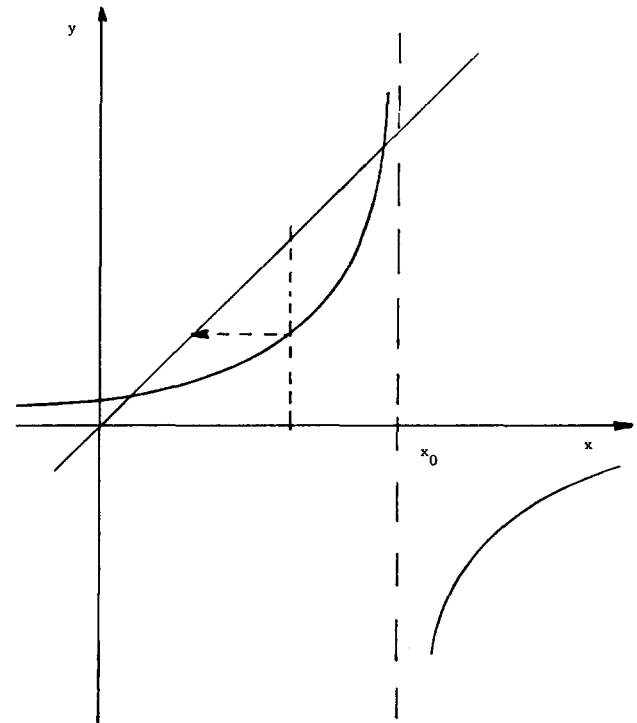


FIG. 1. Geometric proof of the continued-fractional convergence. We may read  $y = f_n$  or  $g_m$ ,  $x = f_{n+1}$  or  $g_{m+1}$  and  $x_0 = 4n^2/\lambda^2$  or  $m/\mu$ , respectively.

Obviously, the unique values of  $f_n$  and  $g_m$  will be obtained in the infinite-dimensional limit  $m, n \rightarrow \infty$ . Q.E.D.

We may conclude that the continued-fractional condition (3.8) extends the Hill-determinant requirement (3.3) to the limit  $M, N \rightarrow \infty$ . In this sense, it represents a tentative analytic specification of energies, analogous to the sextic anharmonic-oscillator conjecture of Singh *et al.*<sup>3</sup> In the latter case, a partial proof<sup>10</sup> and partial disproof<sup>11</sup> of the conjecture became available during the further development. In the forthcoming sections, a similar reinterpretation and a rigorous background will also be given to the present "eigenvalue" condition (3.8).

#### IV. A RIGOROUS INTERPRETATION OF THE HILL-DETERMINANT ZEROS

##### A. The asymptotic behavior of the difference Schrödinger equation

Let us consider the Schrödinger equation (3.2) in the  $|n| \gg 1$  asymptotic region where it acquires a particularly simple form,

$$\mu p_n - (n+1)p_{n+1} + (\lambda^2/4n)p_{n+2} = 0, \quad |n| \gg 1. \quad (4.1)$$

First, let us choose positive  $n \gg 1$  and notice that an apparently "dominant" part of Eq. (4.1),

$$\mu \tilde{p}_n - (n+1)\tilde{p}_{n+1} = 0, \quad n \gg 1, \quad (4.2)$$

has an exact solution

$$\tilde{p}_n = (\mu^n/n!)p_0. \quad (4.3)$$

In the next step, we may change the variables. Denote

$$p_n = (\mu^n/n!) q_n^{(+)}, \quad n > 0, \quad (4.4)$$

and convert full Eq. (4.1) into an equivalent difference equation

$$q_{n+1}^{(+)} - q_n^{(+)} = [\lambda^2\mu/4n(n+1)(n+2)] q_{n+2}^{(+)}. \quad (4.5)$$

Here, we may try to decompose  $q_n^{(+)}$  into a Taylor series

$$q_{n+k}^{(+)} = q_n^{(+)} + k \frac{d}{dn} q_n^{(+)} + \dots,$$

and obtain an approximate differential equation in the lowest nontrivial order,

$$\frac{d}{dn} \tilde{q}_n^{(+)} = \frac{\lambda^2\mu}{4n^3} \tilde{q}_n^{(+)}. \quad (4.6)$$

Its solution

$$q_n^{(+)} \approx \tilde{q}_n^{(+)} = \exp(-\lambda^2\mu/8n^2), \quad n \gg 1, \quad (4.7)$$

improves the leading-order estimate (4.3) and shows the extremely weak  $n$  dependence of the product  $p_n \cdot n!\mu^{-n}$  for large indices  $n \gg 1$ .

Let us now alternatively take Eq. (4.1) as a recurrent definition of  $p_{n+2} = p_{n+2}(p_{n+1}, p_n)$ . Obviously, its latter argument (the contribution of  $p_n$ ) may also become negligible. In this setting, the relation

$$-(n+1)\tilde{p}_{n+1} + (\lambda^2/4n)\tilde{p}_{n+2} = 0, \quad n \gg 1, \quad (4.8)$$

is a counterpart to Eq. (4.2). Let us modify the change of variables (4.4),

$$p_n = \lambda^{-2n} 4^n (n-1)!(n-2)! \hat{q}_n^{(+)}, \quad n > 1, \quad (4.9)$$

and replace Eq. (4.5) by a new reparametrization of Eq. (4.1),

$$\hat{q}_{n+2}^{(+)} - \hat{q}_{n+1}^{(+)} = -[\mu\lambda^2/4n(n^2-1)] \hat{q}_n^{(+)}. \quad (4.10)$$

The latter equation may again be analyzed in the same manner as before. We decompose  $\hat{q}_n^{(+)}$  into a Taylor series and arrive at the second, independent asymptotic solution of our three equivalent difference Eqs. (4.1), (4.5), or (4.10), with

$$\hat{q}_n^{(+)} \approx \hat{\tilde{q}}_n^{(+)} = \exp(\lambda^2\mu/8n^2), \quad n \gg 1. \quad (4.11)$$

Summarizing the whole procedure, we may write the general form of coefficients  $p_n$  ( $n \gg 1$ ) in the form of the superposition  $c_1 \tilde{p}_n + c_2 \hat{\tilde{p}}_n$ ,

$$p_n = c_1 \frac{\mu^n}{n!} \exp\left(-\frac{\lambda^2\mu}{8n^2}\right) + c_2 \frac{4^n(n-1)!(n-2)!}{\lambda^{2n}} \times \exp\left(+\frac{\lambda^2\mu}{8n^2}\right), \quad n \gg 1. \quad (4.12)$$

##### B. Convergence of the Ansatz $\psi(r)$

In the  $n \ll -1$  asymptotic domain of indices, it is easy to repeat all the manipulations of Sec. IV A. Indeed, the modified pair of Ansätze (4.4) and (4.9),

$$p_{-m} = \frac{(-1)^m(m-1)!}{\mu^m} q_m^{(-)} = \frac{\lambda^{2m}}{4^m m!(m+1)!} \hat{q}_m^{(-)}, \quad m > 0, \quad (4.13)$$

leads to the two new (equivalent) forms of Eq. (4.1) with negative subscripts,

$$q_m^{(-)} - q_{m-1}^{(-)} = [\mu\lambda^2/4m(m-1)(m-2)] q_{m-2}^{(-)}, \\ \hat{q}_{m-1}^{(-)} - \hat{q}_{m-2}^{(-)} = -[\mu\lambda^2/4m(m^2-1)] \hat{q}_m^{(-)}. \quad (4.14)$$

The negative-index analog of Eq. (4.12) will read

$$p_{-m} = c_3 \frac{(-1)^m(m-1)!}{\mu^m} \exp\left(-\frac{\lambda^2\mu}{8m^2}\right) + c_4 \frac{\lambda^{2m}}{4^m m!(m+1)!} \exp\left(+\frac{\lambda^2\mu}{8m^2}\right), \\ m \gg 1. \quad (4.15)$$

Thus we arrive at the following result.

**Lemma 2:** A necessary and sufficient condition of convergence of the doubly infinite Laurent series (3.1) may be formulated as a restriction  $c_2 = c_3 = 0$  imposed on the asymptotics (4.12) and (4.15),

$$p_n \approx \begin{cases} \mu^n/n!, & n \gg 1, \\ (\lambda/2)^{2m}/m!(m+1)!, & m = -n \gg 1. \end{cases} \quad (4.16)$$

**Proof:** It is trivial and follows from an immediate application of the standard convergence criteria to (3.1), with

$$\frac{p_n}{p_{n-1}} = f_n = \begin{cases} O(n^2), & c_2 \neq 0, \\ \mu/n + \text{corrections}, & c_2 = 0, \end{cases} \quad n \gg 1, \quad (4.17)$$

and

$$\frac{p_n}{p_{n+1}} = g_m = \begin{cases} O(|n|), & c_3 \neq 0, \\ \lambda^2/4n^2 + \text{corrections}, & c_3 = 0, \end{cases} \quad n \ll -1. \quad (4.18)$$

This completes the proof.

An understanding of Eq. (4.16) is a key point of the present construction: Our formal solution (3.1) of the differential Eq. (1.2) exists if and only if the ambiguous solution of Eq. (3.2) is made unique by means of the “mathematical” boundary conditions (4.16).

We may compare (4.17), (4.18), (3.11), and (3.13) and see that the mathematical conditions are satisfied if and only if we pick up the specific, continued-fractional direction of recurrences (3.2) [rewritten as Eqs. (3.6)–(3.8)]. Thus we may conclude that the continued-fractional restriction (3.8) and prescriptions

$$p_n = \begin{cases} p_0 \prod_{k=1}^n f_k, & n > 0, \\ p_0 \prod_{k=1}^m g_k, & n = -m < 0, \end{cases} \quad (4.19)$$

define a convergent (not necessarily normalizable) solution  $\psi(r)$  in (3.1) of our radial Schrödinger Eq. (1.2).

## V. A PHYSICAL SPECIFICATION OF THE PAIR OF PARAMETERS $\kappa$ AND $E$

We have seen that the Hill-determinant condition (3.3) or (3.8) is a necessary and sufficient condition of applicability (convergence) of the power-series expansions (3.1). The analytic continued-fractional form of this equation is quickly convergent [cf. Eqs. (3.11) and (3.13)] and compatible with a truncation of  $\mathcal{H} - M, N$  at rather small dimensions.

Now, on a background of our numerical tests, we conjecture that in a vicinity of the physical energies  $E = E_n^{(\text{phys})}$ ,  $n = 0, 1, \dots$ , the continued-fractional equation (3.8) will normally have two roots  $\kappa = \kappa_1(E)$  and  $\kappa = \kappa_2(E)$  and that the two corresponding solutions  $\psi^{(\kappa)}(r)$  (3.1) of Eq. (1.2) will be linearly independent. For the particular choices of the couplings we have made, the roots  $\kappa_1$  and  $\kappa_2$  always formed a complex conjugate pair.

We may here recall that for  $\mu = 0$ , a similar rule also holds: the Hill-determinant condition of convergence of the Laurent series (3.1) gives the imaginary pair of roots  $\kappa_{1,2}$ .<sup>4</sup> In particular, for  $r \gg 1$  and  $\mu \neq 0$  where wave functions closely resemble the freely traveling waves, a number of formal analogies with the Mathieu functions<sup>4</sup> still may be recovered in the present case: We omit any mathematical details of this type here.

In a vicinity of an exact binding energy  $E_n^{(\text{phys})}$ , the wave function acquires a new node (zero) in such a way that<sup>12</sup> the number  $m$  of these nodes  $\tilde{r}_1, \tilde{r}_2, \dots, \tilde{r}_m$  is equal to  $n$  for  $E_n^{(\text{phys})} < E < E_{n+1}^{(\text{phys})}$  and  $n = 0, 1, \dots$ . In the regular-potential case, merely the limiting transition  $r \rightarrow 0$  is simplified. Thus we

may consider a general solution of Eq. (1.2),

$$\psi(r) = d_1 \psi^{(\kappa_1)}(r) + d_2 \psi^{(\kappa_2)}(r), \quad d_1^2 + d_2^2 = 1; \quad (5.1)$$

and (a) requiring that  $\psi(r)$  in (5.1) satisfies the  $r \rightarrow 0$  boundary conditions, specify the so-called regular solution<sup>1</sup>

$$\begin{aligned} \psi_R(r) &= d_{1R} \psi^{(\kappa_1)}(r) + d_{2R} \psi^{(\kappa_2)}(r), \\ d_{iR} &= \lim_{\tilde{r}_i \rightarrow 0} d_i(\tilde{r}_i), \quad i = 1, 2, \\ d_1(\tilde{r}_1) \psi^{(\kappa_1)}(\tilde{r}_1) &= -d_2(\tilde{r}_1) \psi^{(\kappa_2)}(\tilde{r}_1); \end{aligned} \quad (5.2)$$

(b) requiring that  $\psi(r)$  in (5.1) satisfies the  $r \rightarrow \infty$  boundary condition, specify another unphysical, so-called Jost solution<sup>1</sup>

$$\begin{aligned} \psi_J(r) &= d_{1J} \psi^{(\kappa_1)}(r) + d_{2J} \psi^{(\kappa_2)}(r), \\ d_{iJ} &= \lim_{\tilde{r}_i \rightarrow \infty} d_i(\tilde{r}_i), \quad i = 1, 2; \end{aligned} \quad (5.3)$$

and (c) matching the logarithmic derivatives of (5.2) and (5.3) at an arbitrary  $r \in (0, \infty)$ , we may determine  $E = E^{(\text{phys})}$  and the physical solution  $\psi^{(\text{phys})}(r) = \psi_R(r) = \psi_J(r)$ .

As well as in the other cases solvable by means of the convergent power-series expansions, the matching condition is most easily formulated either at  $r \approx \infty$  or at  $r \approx 0$ . In both cases, one of the solution [ $\psi_R$  or  $\psi_J$ , i.e., (5.2) or (5.3)] is needed and the respective relation

$$\psi_{R,J}(r) \approx 0 \quad (5.4)$$

may be interpreted as an appearance of a new node in  $\psi(r)$ . It is taking place slightly above the exact energy level, so that an efficient and reliable numerical algorithm results.

In both the regular and Jost cases, the physical boundary conditions may be understood as a simultaneous change of sign of  $\psi(r)$  in (5.1) at a pair of points  $r \approx 0$  and  $r \approx \infty$ . Then, the related conditions are

$$\begin{aligned} \tilde{d}_1 \psi^{(\kappa_1)}(r_0) + \tilde{d}_2 \psi^{(\kappa_2)}(r_0) &= 0, \quad r_0 \ll 1, \\ \tilde{d}_1 \psi^{(\kappa_1)}(r_1) + \tilde{d}_2 \psi^{(\kappa_2)}(r_1) &= 0, \quad r_1 \gg 1, \end{aligned} \quad (5.5)$$

and specify the unknowns. The mathematical Hill-determinant requirement (3.8) must be complemented by the physical, two-dimensional eigenvalue condition

$$\det \begin{pmatrix} \psi^{(\kappa_1)}(r_0) & \psi^{(\kappa_2)}(r_0) \\ \psi^{(\kappa_1)}(r_1) & \psi^{(\kappa_2)}(r_1) \end{pmatrix} = 0, \quad r_0 \ll 1, \quad r_1 \gg 1. \quad (5.6)$$

This is our final result.

## VI. NUMERICAL TESTS

Our potential (1.1) is a smooth function of  $r$  and also the threshold and asymptotic behavior of the wave function is well known [cf. Eqs. (2.1) and (2.2), respectively]. As a consequence, the radial Schrödinger equation (1.2) may easily be solved, say, by the standard Runge–Kutta method.<sup>13</sup> A sample of the resulting energies may be found here in Table I. They are compatible also with the variational results of Ref. 5 where  $E_0 = 4.031\,97$  has been obtained for the same force.

Even in the purely numerical context, our present con-

TABLE I. The first two “exact” energies as obtained by the Runge–Kutta method for  $\mu^2 = 1$  and  $\lambda^2 = 0.4$ .

| Matching points |       | Energies    |              |
|-----------------|-------|-------------|--------------|
| $r_0$           | $r_1$ | $E^{(0)}$   | $E^{(1)}$    |
| 0.045           | 3.5   | 4.0321 9    | 8.342 757    |
| 0.040           | 4.0   | 4.0319 785  | 8.316 366 0  |
| 0.035           | 4.5   | 0.0319 7166 | 8.314 624 24 |
| 0.030           | 5.0   | 4.0319 7139 | 8.314 565 23 |

struction of wave functions is “almost non-numerical” and may be understood as a specific generalization of the Mathieu functions<sup>4</sup> (where similar *Ansätze* are used). Of course, the determination of energies remains purely numerical in practice. Still, the purely analytic and everywhere convergent character of  $\psi$ 's may remain useful. Without going into detail, we would like to point out only that an improvement of the matching conditions (5.5) (with derivatives) becomes extremely simple now (a more detailed analysis of this technicality will be described elsewhere<sup>14</sup>).

Methodically, an important feature of our present example may be seen in the presence of the two free parameters ( $E$  and  $\kappa$ ). This forces us to complement the single Hill-determinant condition (as used, e.g., in Ref. 3) by an additional equation [cf. (5.6)]. Of course, similar additional conditions may be expected to appear also in the more complicated systems. In a way, our present analysis indicates their possible treatment by the same techniques. Indeed, the computer implementation of our approach remains extremely easy: The continued fractions converge quickly. In fact, there is no need to use the acceleration of convergence via Eqs. (3.12) or (3.13). A full double-precision compatibility between the  $M, N = 100$  and 800 results has been observed in our tests.

In Table II, a sample of results of the present method deviates from the “exact” results of Table I by the rounding errors only. This confirms that the implementation [and, in particular, a search for the complex roots of Eqs. (3.8)] of our prescription is easy and does not lead to any problems in principle. The tests confirm the good convergence properties derived above by purely algebraic means.

## VII. DISCUSSION

At each energy value  $E$ , our Laurent-series *Ansatz* becomes convergent (solves the Schrödinger equation) for a pair of parameters  $\kappa_1$  and  $\kappa_2$  specified by the continued-fractional condition. The physical binding energies then become determined, in a more or less standard way, from the condition  $\|\psi\| < \infty$ .

Methodically, our superposition of the two exactly solv-

TABLE II. The energies and  $\kappa$ 's obtained by the present method.

| Matching points |       | Roots of Eq. (3.8) |                      | Energies     |
|-----------------|-------|--------------------|----------------------|--------------|
| $r_0$           | $r_1$ | Re $\kappa_{1,2}$  | Im $\kappa_{1,2}$    | $E$          |
| 0.045           | 3.5   | 0.5                | $\pm 0.606\ 102$     | 4.032 19     |
| 0.040           | 4.0   | 0.5                | $\pm 0.606\ 083\ 7$  | 4.031 978 4  |
| 0.035           | 4.5   | 0.5                | $\pm 0.606\ 083\ 08$ | 4.031 971 69 |
| 0.030           | 5.0   | 0.5                | $\pm 0.606\ 083\ 10$ | 4.031 971 34 |
| 0.045           | 3.5   | 0.5                | $\pm 0.858\ 110$     | 8.342 755    |
| 0.040           | 4.0   | 0.5                | $\pm 0.857\ 011$     | 8.316 365 4  |
| 0.035           | 4.5   | 0.5                | $\pm 0.856\ 938\ 7$  | 8.314 624 04 |
| 0.030           | 5.0   | 0.5                | $\pm 0.856\ 936\ 31$ | 8.314 565 38 |

able potentials finds a surprisingly natural perturbative interpretation for small  $\mu$  rather than for small  $\lambda$ . The “anharmonicity”  $\lambda r^{-4}$  preserves its nonperturbative character even for small  $\lambda$ .

In the harmonic-oscillator  $\lambda \rightarrow 0$  limit, and irrelevance of the  $n < -1$  part of  $\psi(r)$  ( $g_1 \approx 0$ ) gives  $\kappa(\kappa - 1) \approx l(l + 1)$ , i.e.,  $\kappa_1 \approx l + 1$  and  $\kappa_2 \approx -l$ . In the same approximation, we then get  $\tilde{d}_1 \approx 1$  and  $\tilde{d}_2 \approx 0$ , i.e.,  $\psi^{(\kappa_1)}(r_0) \approx 0$  in our secular equation. This leads to the correct power-series termination requirement and harmonic oscillator spectrum  $E \approx \mu(4N + 2\kappa_1 + 1)$  as it should.

Our example clarifies an overall structure of the so-called Hill-determinant eigenvalue method (cf. Ref. 11 and references given therein). It may give both the physical and unphysical energies and solutions in general, depending on the structure of the underlying *Ansatz*. Our present analysis illustrates this and how the corresponding proofs may be based on an asymptotic solution of the related difference equations.

<sup>1</sup>R. G. Newton, *Scattering Theory of Waves and Particles* (McGraw-Hill, New York, 1982).

<sup>2</sup>C. Itzykson and J. B. Zuber, *Quantum Field Theory* (McGraw-Hill, New York, 1980).

<sup>3</sup>V. Singh, S. N. Biswas, and K. Datta, *Phys. Rev. D* **18**, 1901 (1978).

<sup>4</sup>H. Bateman and A. Erdelyi, *Higher Transcendental Functions III* (McGraw-Hill, New York, 1955).

<sup>5</sup>M. Znojil, *Phys. Lett. A* **101**, 66 (1984).

<sup>6</sup>M. Znojil, *J. Math. Phys.* **25**, 2979 (1984).

<sup>7</sup>C. Lanczos, *J. Res. Natl. Bur. Stand.* **45**, 255 (1950).

<sup>8</sup>J. H. Wilkinson, *The Algebraic Eigenvalue Problem* (Clarendon, Oxford, 1965).

<sup>9</sup>H. S. Wall, *Analytic Theory of Continued Fractions* (Van Nostrand, New York, 1943).

<sup>10</sup>M. Znojil, *Phys. Rev. D* **26**, 3730 (1982).

<sup>11</sup>R. N. Chaudhuri, *Phys. Rev. D* **31**, 2687 (1985).

<sup>12</sup>E. L. Ince, *Ordinary Differential Equations* (Dover, New York, 1956).

<sup>13</sup>W. E. Milne, *Numerical Solutions of Differential Equations* (Wiley, New York, 1953).

<sup>14</sup>M. Znojil, “The analytic continued fractions and bound states in the potential  $\mu^2 r^2 + \lambda^2 (1/r)^4$ ,” Univ. Kaiserslautern preprint, 1988.